

Forecasting Monthly Median Dwelling Prices in Porto: Smoothing, Decomposition and ARIMA Methods

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Forecasting Methods and Time Series

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Abstract

This report forecasts monthly median dwelling prices in Porto (EUR/m²) using multiplicative smoothing and decomposition methods, together with an ARIMA model fitted on a log-transformed, differenced series. The sample spans January 2011–February 2026 ($n = 182$). A hold-out test set of 24 months (March 2024–February 2026) is used to compare methods. ETS achieves the lowest test RMSE (326 EUR/m²); all methods under-forecast the accelerated growth in the test period. Out-of-sample 95% prediction intervals for the next 24 months are reported for each approach.

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1 Introduction and data description

1.1 Objective

The objective is to forecast a real monthly time series using smoothing, decomposition, and ARIMA/SARIMA methods, to compare their predictive performance on a test set, and to produce out-of-sample forecasts with 95% prediction intervals.

1.2 Data source

The variable is the monthly median price per square metre (EUR/m²) for dwellings in Porto (series code 11A1312: Porto). Data are in `data/timeSeriesPorto.csv` and cover January 2011–February 2026.

1.3 Features of the series

Figure 1 shows a long upward trend: prices remain near 1 000 EUR/m² until about 2015, then increase steadily with faster growth after 2020. A mild seasonal pattern is visible; the seasonal boxplot (Figure 2) confirms that within-year variation is small relative to the trend. A spike around 2022–2023 is followed by a correction and renewed growth.

The ACF and PACF of the levels (Figure 3) decay slowly with spikes at lags 12, 24 and 36, indicating non-stationarity and annual seasonality. Following the course guidelines, unit-root tests designed for non-seasonal series (ADF, PP, KPSS) are not applied; stationarity is assessed through visual inspection, ACF/PACF, and differencing.

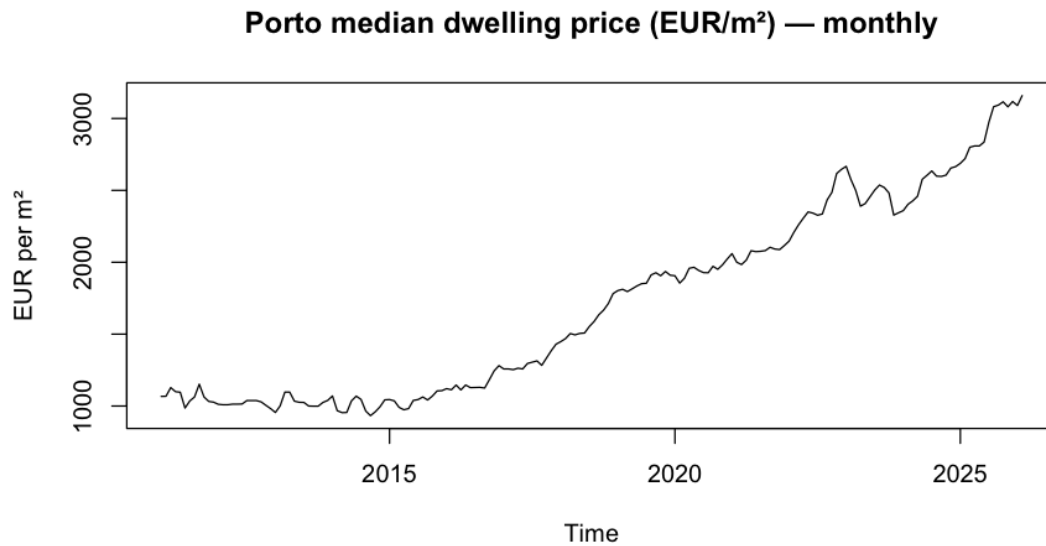


Figure 1: Monthly median dwelling price in Porto (EUR/m²).

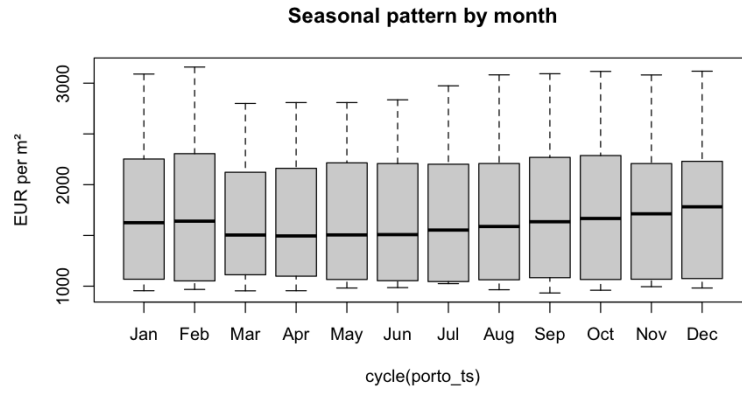


Figure 2: Seasonal boxplot by calendar month.

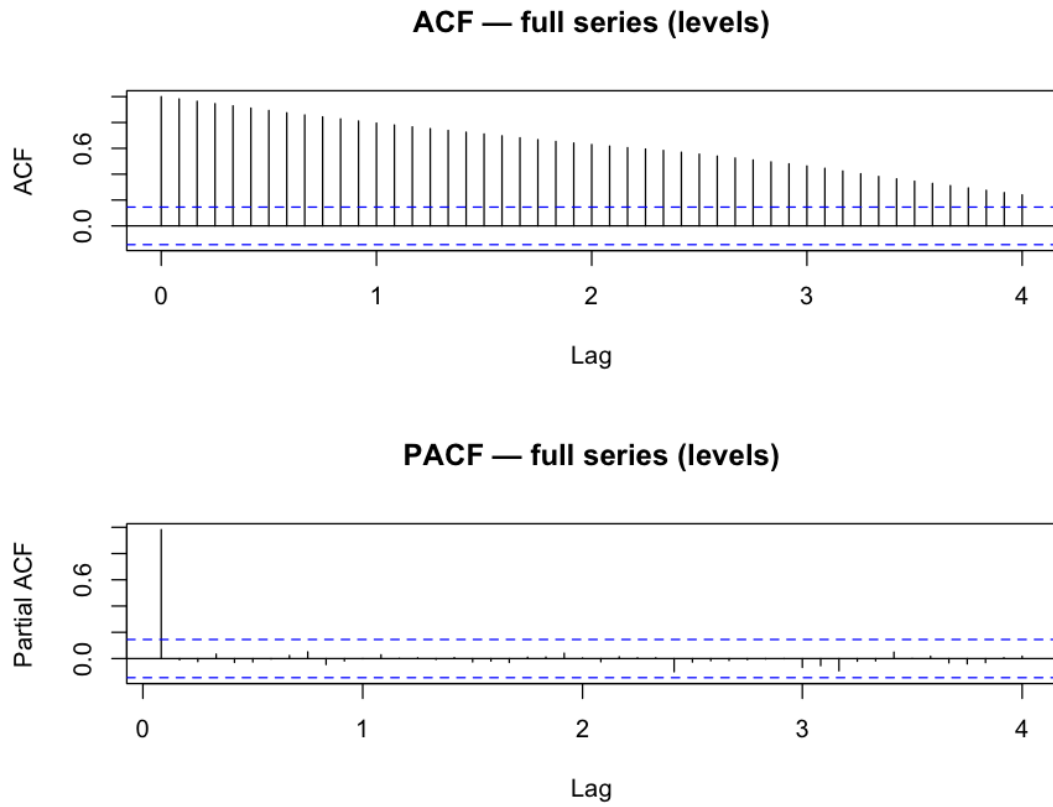


Figure 3: ACF and PACF of the series in levels.

1.4 Train/test split

The last $h = 24$ months (March 2024–February 2026) form the test set. All models are estimated on the training sample (January 2011–February 2024).

2 Decomposition methods

2.1 Multiplicative decomposition

Because seasonal effects are more naturally expressed as proportional to the level when a strong trend is present, **multiplicative** decomposition is used:

$$y_t = T_t \times S_t \times R_t,$$

where T_t is trend-cycle, S_t seasonal, and R_t remainder.

Figure 4 shows the classical multiplicative decomposition. Seasonal factors fluctuate around 1.0 with a narrow range (roughly 0.98–1.05), confirming that seasonality is small in relative terms but non-negligible in theory.

STL is applied to $\log(y_t)$ so that additive components on the log scale correspond to multiplicative structure on the original scale (Figure 5). The seasonally adjusted series is obtained as $y_t^{\text{SA}} = y_t / S_t$ (Figure 6).

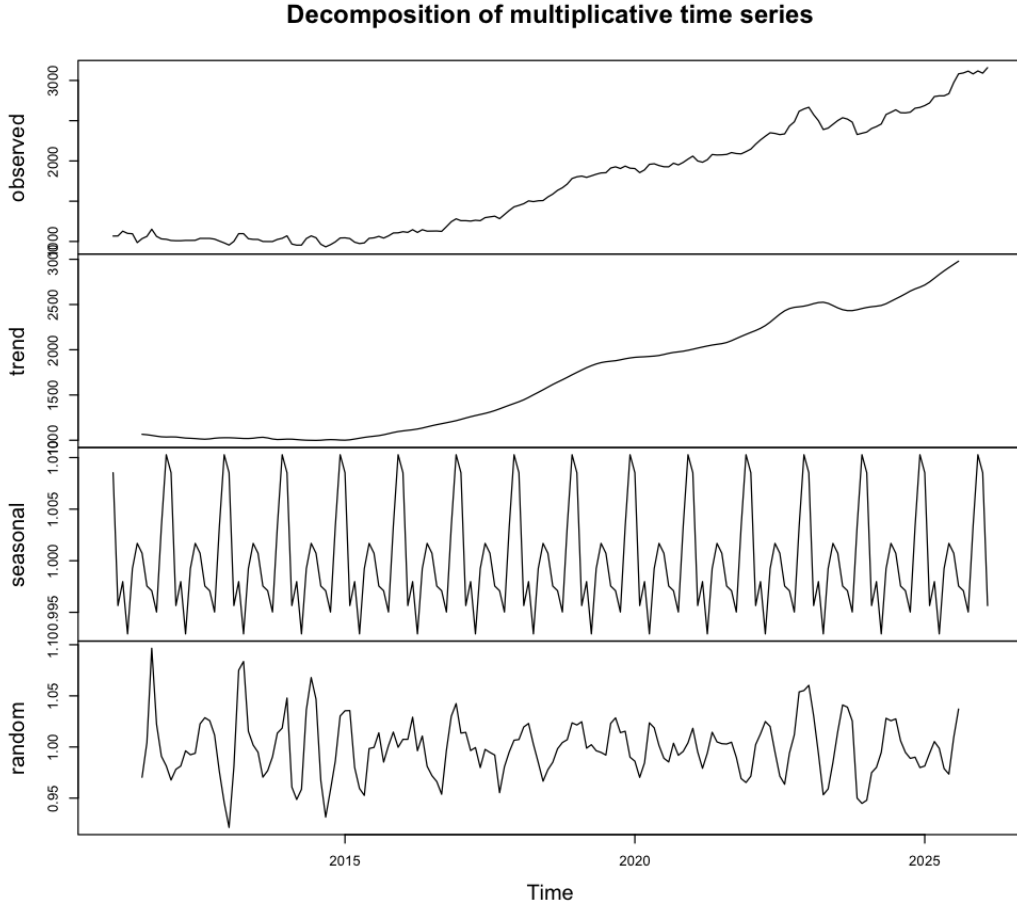


Figure 4: Classical multiplicative decomposition.

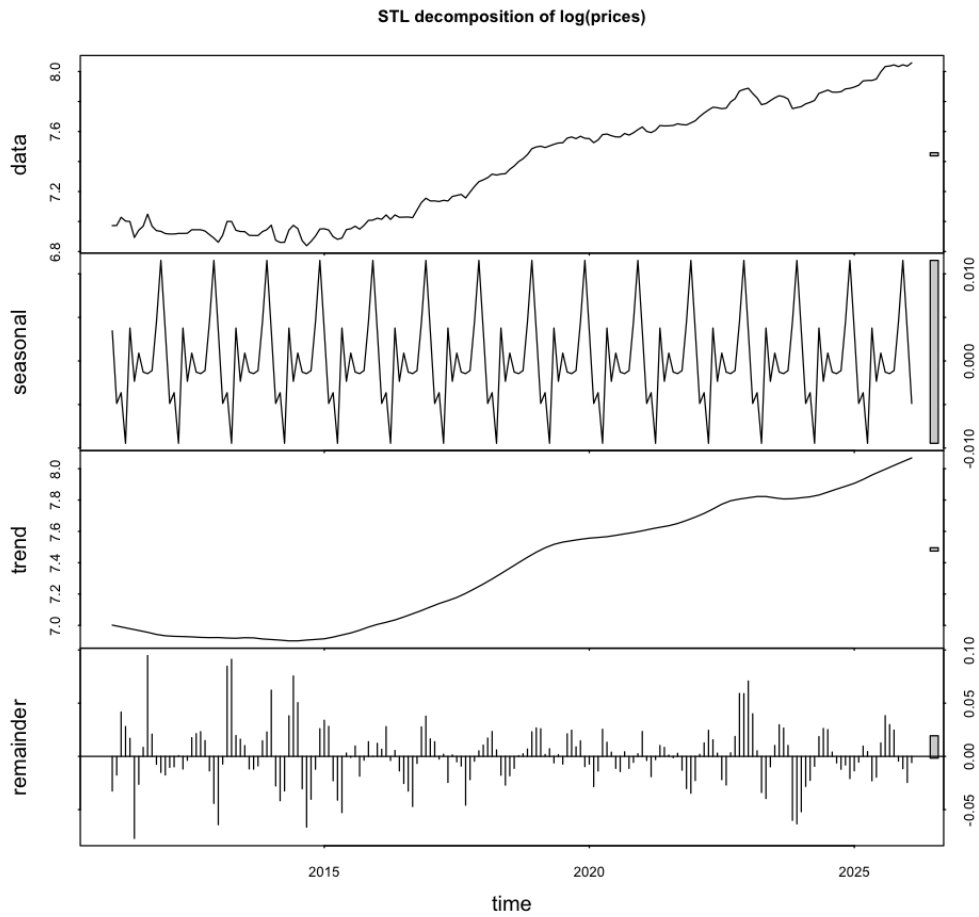


Figure 5: STL decomposition of $\log(y_t)$.

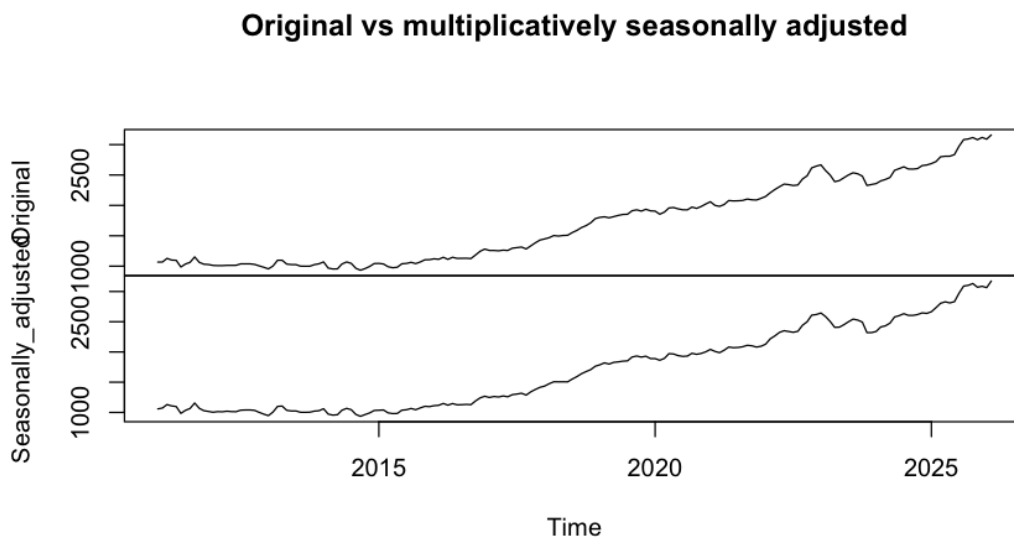


Figure 6: Original vs. multiplicatively seasonally adjusted series.

3 Smoothing methods

3.1 Holt-Winters (multiplicative)

Holt-Winters with **multiplicative seasonality** is fitted to the training sample. Estimated smoothing parameters are $\alpha = 0.804$, $\beta = 0.018$, and $\gamma = 1.000$. Seasonal factors s_m range from about 0.983 (February) to 1.047 (October). A damped-trend variant is also considered.

3.2 ETS and STLF

The `ets()` function selects **ETS(A,A,N)** on the training window (additive trend, no seasonal term; AIC = 1981.5). STLF combines STL with ETS forecasting on the seasonally adjusted component.

Figures 7–9 show each method against the test set with 95% prediction intervals.

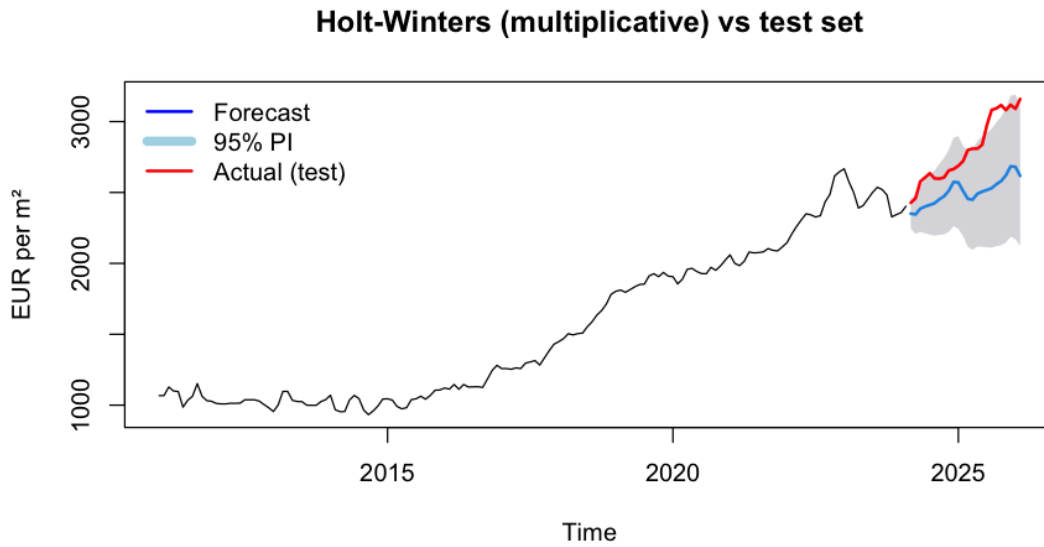


Figure 7: Holt-Winters (multiplicative) vs. test set.

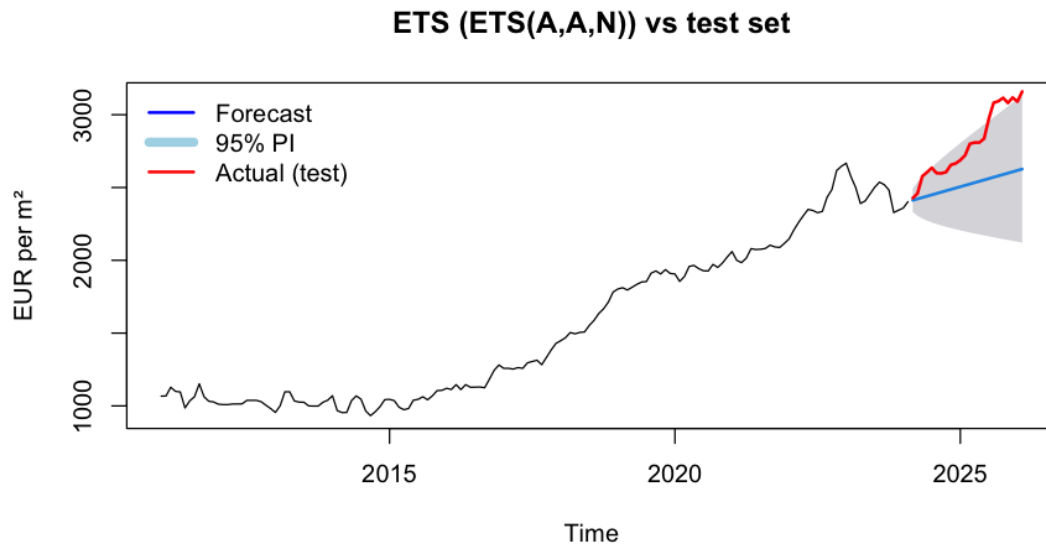


Figure 8: ETS vs. test set.

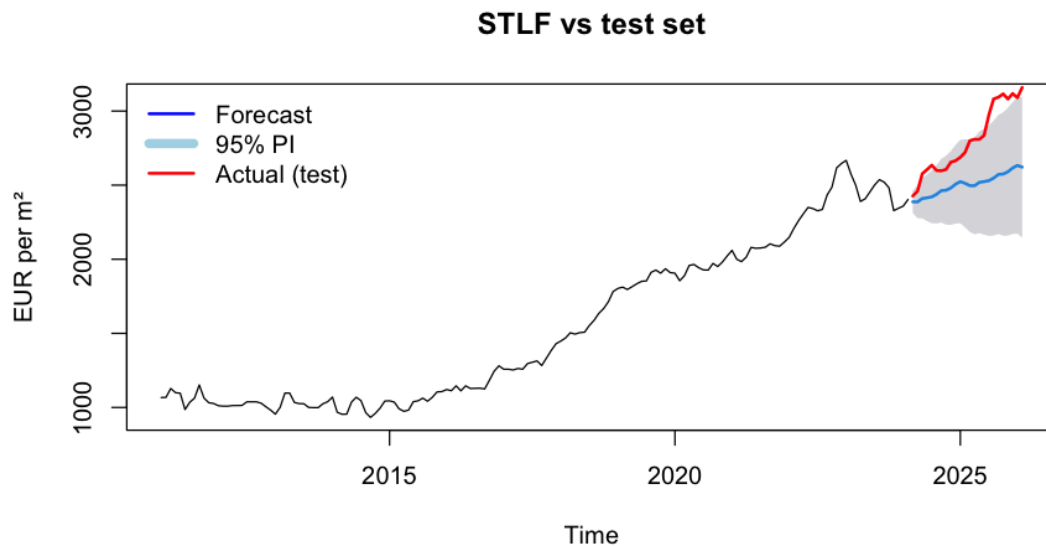


Figure 9: STL vs. test set.

4 ARIMA modelling

4.1 Transformations and differencing

Prices grow over time and variance increases with the level. A **log transformation** stabilises variance; **first differencing** of $\log(y_t)$ removes the trend and yields a series that is approximately stationary (Figures 12–10). The ACF/PACF of the differenced log series (Figure 11) guide model identification.

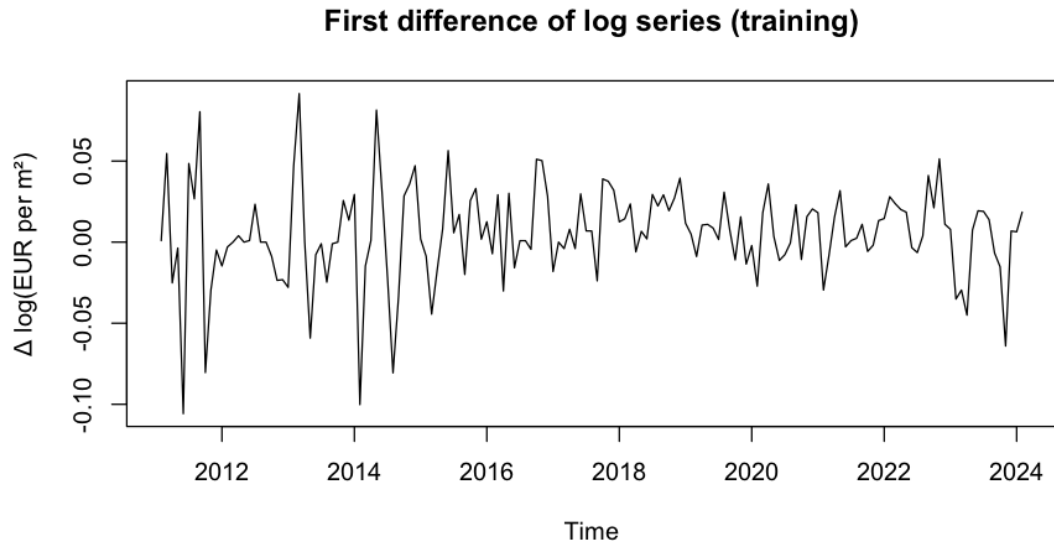


Figure 10: First difference of $\log(y_t)$ on the training sample.

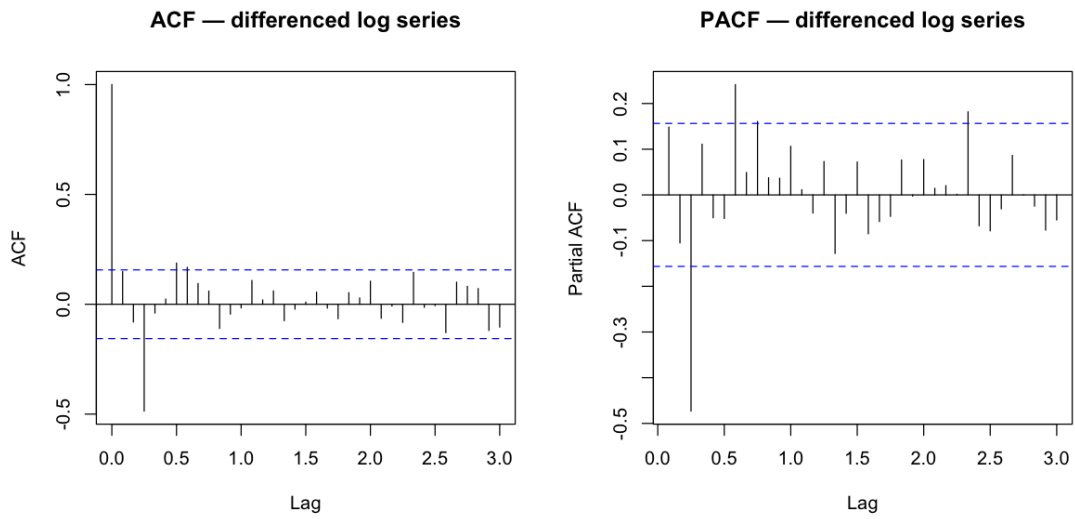


Figure 11: ACF and PACF of the differenced log series.

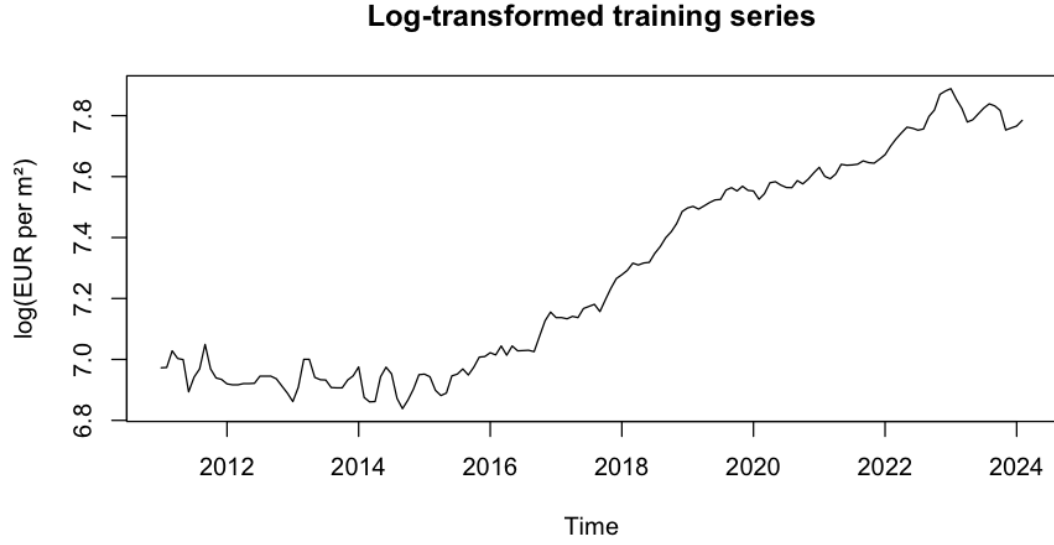


Figure 12: Log-transformed training series.

4.2 Model short-list and AIC

Three candidates are compared on the training sample (all with $d = 1$ on the log scale):

- **Candidate A:** `auto.arima` with `seasonal = FALSE`
- **Candidate B:** `auto.arima` with `seasonal = TRUE`
- **Candidate C:** manual SARIMA(3, 1, 0)(0, 0, 1)₁₂

Both auto-selected models yield **ARIMA**(1, 1, 4) ($\text{AIC} = -710.4$); the manual SARIMA has higher AIC ($= -693.4$). ARIMA(1, 1, 4) is retained for diagnostics and forecasting. Forecasts on the original scale are obtained by exponentiating the log-scale predictions.

4.3 Diagnostic checking

Figure 13 shows residual diagnostics. The Ljung-Box test on innovation residuals at lag 24 with `fitdf = 5` (estimated AR/MA coefficients) gives $Q^* = 16.80$, $\text{df} = 19$, $p = 0.60$, so residual autocorrelation is not significant at conventional levels.

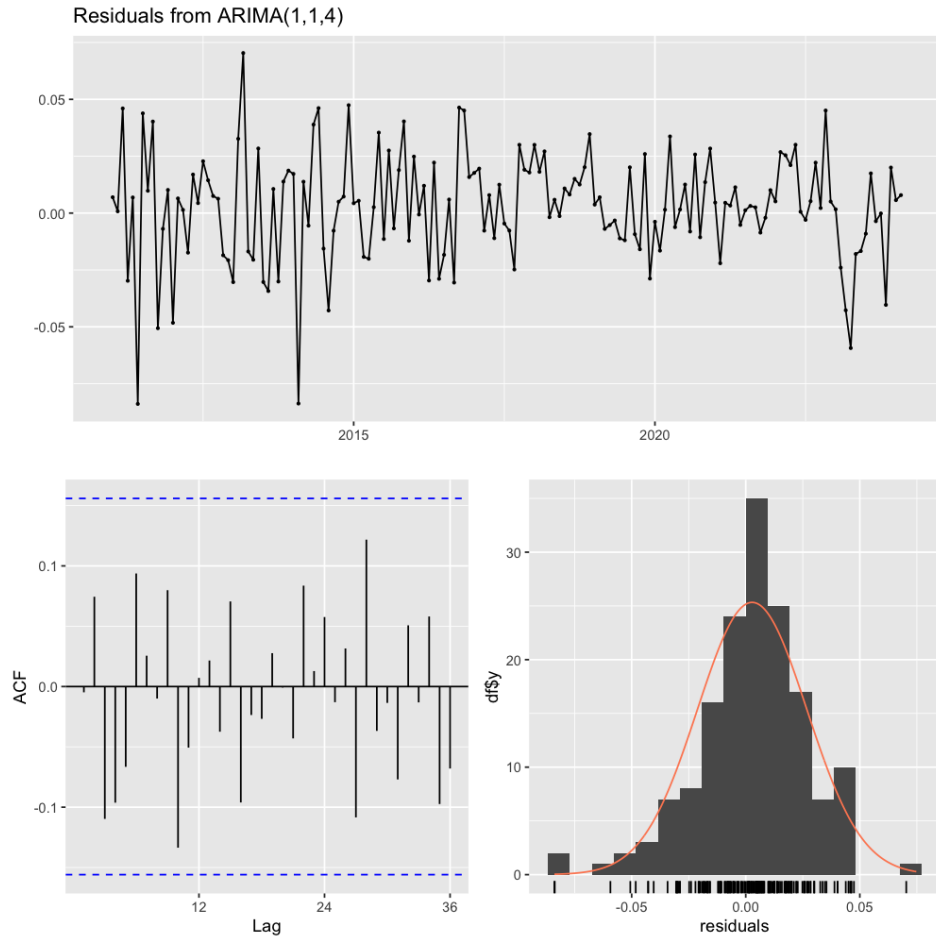


Figure 13: Residual diagnostics for ARIMA(1, 1, 4) on $\log(y_t)$.

Figure 14 compares the back-transformed forecast with the test set in EUR/m².

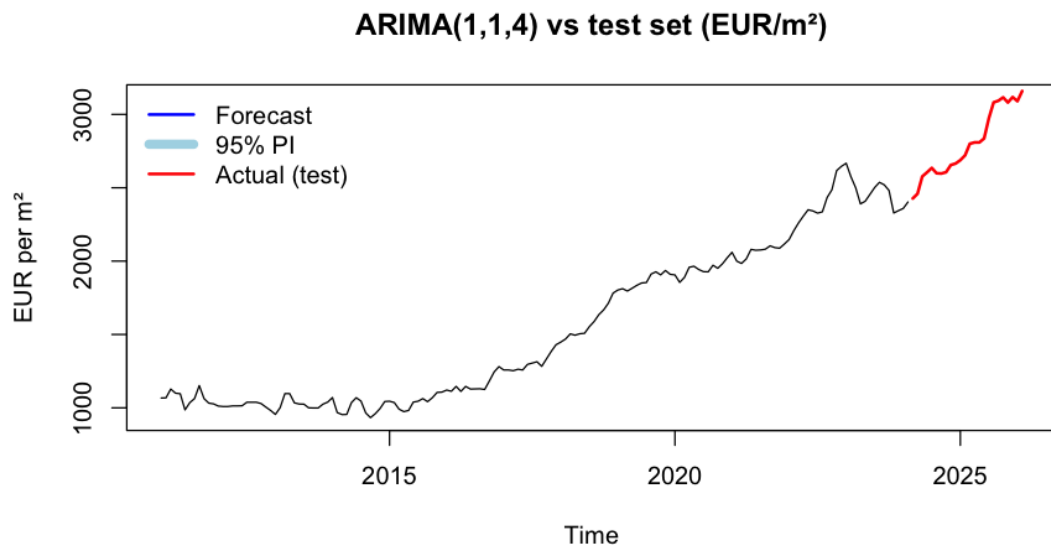


Figure 14: ARIMA(1, 1, 4) forecast vs. test set (EUR/m²).

5 Forecast comparison

Table 1 summarises test-set accuracy. **ETS** achieves the lowest RMSE (326 EUR/m²) and MAPE (9.6%). Holt-Winters (multiplicative) and STLTF perform similarly (RMSE $\approx$ 336). The damped Holt-Winters variant and the log-ARIMA model perform worse on this 24-month test window, with ARIMA showing the largest systematic under-forecast (MAPE 18.3%).

All methods produce positive mean error: actual prices in March 2024–February 2026 rise faster than any model extrapolates from the training history. The 95% intervals generally remain wide enough to contain much of the test path.

Table 1: Test-set accuracy (24-month hold-out).

Method	RMSE	MAE	MAPE (%)	ME
ETS	325.7	280.6	9.62	280.6
HW (mult.)	335.9	295.9	10.20	295.9
STLTF	336.3	294.5	10.13	294.5
HW (damped)	415.4	359.3	12.32	359.3
ARIMA(1, 1, 4) log	582.5	529.0	18.31	529.0

6 Out-of-sample forecasts

Each method is refitted on the full sample and used to forecast the next 24 months (March 2026–February 2028) with 95% prediction intervals (Figures 15–16). Full numerical output is in `outputs/tables/out_of_sample_forecasts_95.csv`.

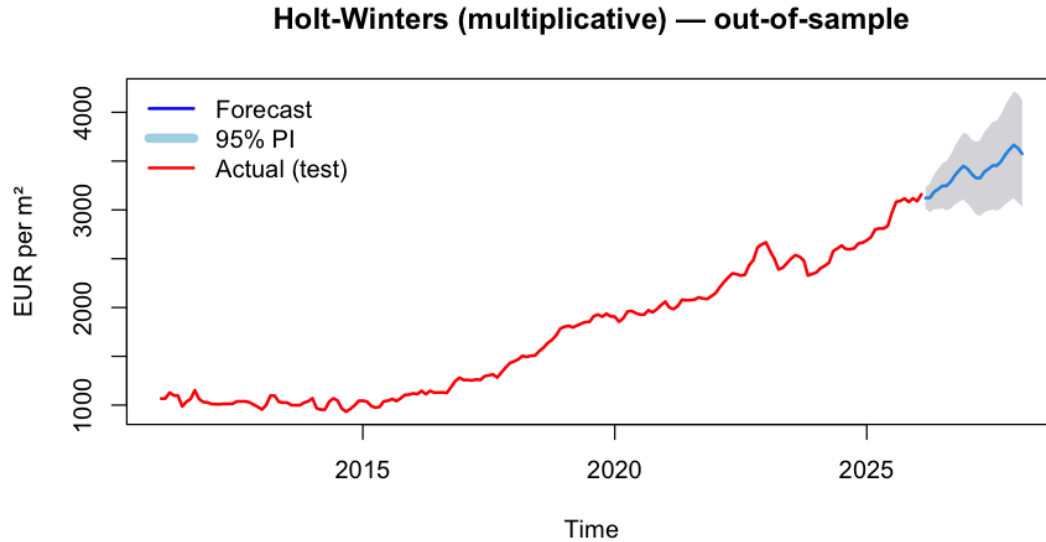


Figure 15: Holt-Winters (multiplicative) — out-of-sample forecast.

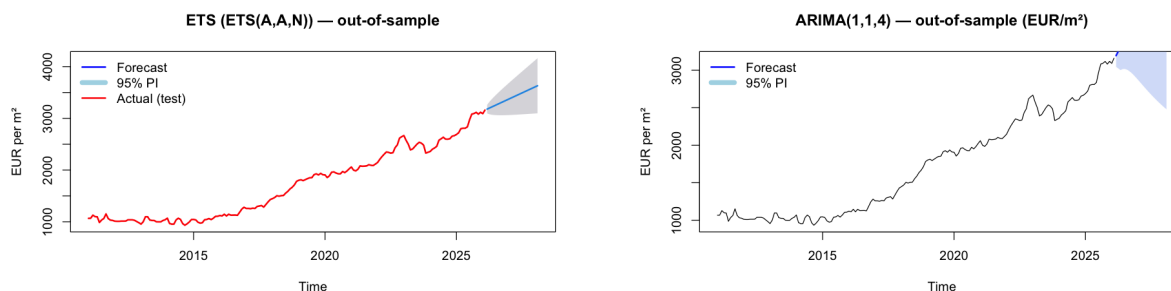


Figure 16: Out-of-sample forecasts: ETS (left) and ARIMA(1, 1, 4) (right).

7 Benefits and limitations

- **Multiplicative decomposition** respects proportional seasonality when the level trends strongly; for this series the seasonal effect is small but the choice is theoretically appropriate.
- **Holt-Winters (multiplicative)** captures trend and seasonality directly; performance is competitive though not best on the test set.
- **ETS** provides automatic model selection and achieves the lowest test RMSE here, but drops explicit seasonality on the training window.
- **STLF** combines robust STL with ETS; results are similar to Holt-Winters.
- **Log-ARIMA(1, 1, 4)** follows standard stationarity practice (log + differencing) and passes residual diagnostics, but under-performs on the test set when growth accelerates; back-transformed intervals assume log-normal innovations.
- **General limitation:** all methods assume historical patterns continue; none fully captures the 2024–2026 acceleration in prices.

References

- Hyndman, R.J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts.
- Portuguese housing price statistics (series 11A1312: Porto), `data/timeSeriesPorto.csv`.